

The dm^3 Criticality Principle: Curvature, Generative Dynamics, and the Dimension Three Spine

Pablo Nogueira Grossi

April 8, 2026

Abstract

We formulate and explore the dm^3 *Criticality Principle*, an abstract generative principle asserting that there is a distinguished curvature coefficient / effective dimension 3 at which dynamics transition from rigid or diffusive behavior to genuinely generative yet controllable behavior. In the dm^3 framework, a generative cycle is decomposed into operators

$$C \rightarrow K \rightarrow F \rightarrow U \quad (\text{Compression} \rightarrow \text{Curvature} \rightarrow \text{Fold} \rightarrow \text{Unfolding}),$$

closed by an entropic boundary operator E .

We show how this principle unifies several a priori distinct domains: Ricci flow and Navier–Stokes in three dimensions, Takeya phenomena in $\mathbb{R}^3$, quantum directional coherence, and discrete linear recurrences (Fibonacci, Tribonacci, Tetranacci, Pentanacci) as a curvature ladder. In each case, the value 3 emerges as the minimal setting where curvature, flow, and topology can fold without trivial collapse or runaway divergence. We present the principle as a conjectural structural axiom, discuss its instantiations, and outline a program for formal verification in a proof assistant.

Contents

1	Introduction	2
2	The dm^3 generative architecture	2
2.1	The operator cycle $C \rightarrow K \rightarrow F \rightarrow U \rightarrow E$	2
2.2	Criticality via a normalized curvature potential	3
3	The dm^3 Criticality Principle	3
3.1	Statement	3
4	Continuous domains at criticality: Ricci and Navier–Stokes	4
4.1	Ricci flow in dimension three	4
4.2	Navier–Stokes in dimension three	4
5	Takeya and directional criticality in $\mathbb{R}^3$	4
5.1	Takeya sets as maximal directional compression	4
5.2	Takeya in quantum mechanics	5

6	Discrete domains: the recurrence ladder	5
6.1	Fibonacci, Tribonacci, Tetranacci, Pentanacci	5
6.2	dm^3 interpretation	5
6.3	The curvature ladder diagram	6
7	Entropic boundary principle	6
7.1	Three layers of entropy	6
8	Program for formalization	6
9	Conclusion	7

1 Introduction

The central claim of this paper is that the number 3 plays a structurally distinguished role in a wide class of dynamical systems, not as numerology but as a *critical curvature dimension* in a generative architecture we call dm^3 . The architecture decomposes dynamics into a cycle

$$C \rightarrow K \rightarrow F \rightarrow U \rightarrow E,$$

where C is compression, K is curvature, F is a fold (a critical transition), U is unfolding or stabilization, and E is an entropic boundary operator that enforces closure.

We formulate a dm^3 *Criticality Principle*: there exists a critical coefficient / dimension 3 at which this cycle becomes nontrivial yet controllable. Below this value, systems are too rigid; above it, they are supercritical and harder to stabilize. We then show how this principle is instantiated in:

- Ricci flow in dimension 3,
- Navier–Stokes in dimension 3,
- Keakeya sets in $\mathbb{R}^3$,
- quantum directional coherence in 3D,
- discrete recurrences: Fibonacci (subcritical), Tribonacci (critical), Tetranacci and Pentanacci (supercritical).

The goal is not to prove new theorems in each domain, but to isolate a common structural spine and state it as a conjectural principle amenable to formalization.

2 The dm^3 generative architecture

2.1 The operator cycle $C \rightarrow K \rightarrow F \rightarrow U \rightarrow E$

We work at an abstract level. A dm^3 *generative system* is specified by:

- a state space X ,
- a compression operator C (reducing degrees of freedom),
- a curvature operator K (introducing nonlinearity / deviation),

- a fold operator F (critical transitions, bifurcations, singularities),
- an unfolding operator U (stabilization, selection of coherent branches),
- an entropic boundary operator E (closure, preventing runaway behavior).

Definition 2.1 (dm^3 generative cycle). A dm^3 generative cycle is a composition

$$G = U \circ F \circ K \circ C$$

on a state space X , together with an entropic boundary operator E such that E bounds the long-term behavior of G (e.g. via energy dissipation, dimensional obstruction, or combinatorial constraints).

The guiding idea is that C reduces, K bends, F folds, U stabilizes, and E closes.

2.2 Criticality via a normalized curvature potential

A convenient toy model for curvature is the normalized cubic potential

$$V_c(q) = q^3 - cq,$$

where $c \in \mathbb{R}$ is an effective curvature coefficient. The special value $c = 3$ yields

$$V_3(q) = q^3 - 3q = (q - 1)^2(q + 2),$$

so that $q = 1$ is a double root. This double root is the algebraic avatar of a fold: a critical point where two branches meet.

Definition 2.2 (Double root at an integer fixed point). We say that c has a *double root at the integer fixed point* if $q = 1$ is a double root of V_c , i.e.

$$V_c(1) = 0, \quad V'_c(1) = 0.$$

For $c = 3$, this holds exactly. In the dm^3 picture, this is the minimal curvature coefficient at which a fold can be aligned with an integer fixed point in a controlled way.

3 The dm^3 Criticality Principle

3.1 Statement

We now state the central principle in an abstract form.

Conjecture 3.1 (dm^3 Criticality Principle). *There exists a critical curvature coefficient $c^* = 3$ (equivalently, a critical effective dimension $d^* = 3$) such that:*

- (i) **Subcritical regime** ($c < 3$, $d < 3$): *all realizations of the dm^3 cycle are rigid or diffusive. Curvature is too weak to generate nontrivial folds; trajectories remain globally regular and fold events are trivial.*
- (ii) **Supercritical regime** ($c > 3$, $d > 3$): *all realizations are supercritical. Curvature-driven stretching dominates; fold events proliferate and U fails to restore coherent structure without additional constraints.*

- (iii) **Critical regime** ($c = 3, d = 3$): the system admits nontrivial yet controlled generative cycles. Algebraically, $V_3(q) = (q - 1)^2(q + 2)$ has a double root at $q = 1$, marking the onset of controlled fold formation at an integer fixed point. This is the minimal setting in which the full cycle $C \rightarrow K \rightarrow F \rightarrow U$ can close without trivialization or blow-up, bounded by E .

The rest of the paper is devoted to showing how this principle is instantiated in several domains.

4 Continuous domains at criticality: Ricci and Navier–Stokes

4.1 Ricci flow in dimension three

In geometric analysis, Ricci flow in dimension 3 is the first setting where curvature, topology, and singularity formation interact in a genuinely generative way. In 2D, the flow is too rigid; in higher dimensions, singularities become more complicated.

- **Subcritical** ($d = 2$): curvature evolution is constrained; topology change is limited.
- **Critical** ($d = 3$): neck-pinching singularities and canonical neighborhoods appear; surgery becomes meaningful.
- **Supercritical** ($d > 3$): singularity models and surgery become more complex.

In dm^3 language, $d = 3$ is the first dimension where C (geometric smoothing), K (curvature), F (singularities), U (surgery), and E (topological/energetic bounds) form a nontrivial yet controllable cycle.

4.2 Navier–Stokes in dimension three

For the incompressible Navier–Stokes equations, dimension 3 is similarly critical. In 2D, global regularity is known; in 3D, vorticity stretching becomes active and the Clay problem asks whether solutions remain regular.

- **C (Compression)**: viscous dissipation and the energy identity reduce high-frequency modes.
- **K (Curvature)**: the vorticity stretching term $(\omega \cdot \nabla)u$ is the curvature-like amplifier, absent in 2D.
- **F (Fold)**: potential finite-time singularities correspond to fold events.
- **U (Unfolding)**: partial regularity and possible relaxation to coherent structures.
- **E (Entropy)**: energy dissipation and enstrophy bounds act as entropic boundaries.

Dimension 3 is the minimal setting where this full cycle is nontrivial.

5 Makeya and directional criticality in $\mathbb{R}^3$

5.1 Makeya sets as maximal directional compression

A Makeya set in $\mathbb{R}^3$ is a set of minimal volume that still contains a unit line segment in every direction. The recent result of Wang–Zahl (2025) shows that any such set must have Minkowski (and Hausdorff) dimension exactly 3.

In dm^3 terms:

- **C**: attempt to compress all directions into minimal volume.
- **K**: directional curvature of the sphere of directions.
- **F**: fold obstruction: one cannot reduce dimension below 3 without losing directions.
- **U**: stable configuration at full dimension 3.
- **E**: dimensional lower bound acts as entropic boundary.

Thus $\mathbb{R}^3$ is the minimal dimension where maximal directional compression is possible without losing completeness.

5.2 Kakeya in quantum mechanics

In quantum mechanics, the analogue of a Kakeya set is a state whose momentum-space support covers all directions. The uncertainty principle and decoherence act as entropic boundaries: one cannot compress the position-space support below full 3D volume while preserving directional completeness.

Again, 3 is the minimal dimension where directional curvature (on S^2), interference, and coherence can coexist in a generative yet bounded way.

6 Discrete domains: the recurrence ladder

6.1 Fibonacci, Tribonacci, Tetranacci, Pentanacci

Consider the k -step linear recurrence

$$X_n = X_{n-1} + X_{n-2} + \cdots + X_{n-k},$$

with characteristic polynomial

$$p_k(x) = x^k - x^{k-1} - \cdots - x - 1.$$

Let λ_k be its dominant root.

- $k = 2$ (Fibonacci): quadratic, $\lambda_2 \approx 1.618$.
- $k = 3$ (Tribonacci): cubic, $\lambda_3 \approx 1.839$.
- $k = 4$ (Tetranacci): quartic, $\lambda_4 \approx 1.928$.
- $k = 5$ (Pentanacci): quintic, $\lambda_5 \approx 1.966$.

6.2 dm^3 interpretation

We interpret k as an effective dimension / curvature depth.

- **Subcritical** ($k = 2$): curvature is weak, folds are trivial, dynamics are rigid.
- **Critical** ($k = 3$): cubic curvature yields one dominant mode and two decaying modes; folds are controlled; the integer lattice acts as a comfortable entropic boundary.

- **Mild supercritical** ($k = 4$): quartic curvature amplifies growth; more memory and richer folds; entropy works harder to maintain coherence.
- **Strong supercritical** ($k = 5$): quintic curvature further amplifies growth; transients are longer and more complex; entropic cost is higher.

This yields a discrete version of the criticality ladder.

6.3 The curvature ladder diagram

This diagram mirrors the continuous case: $2 \rightarrow 3 \rightarrow 4+$ as rigid $\rightarrow$ generative $\rightarrow$ supercritical.

7 Entropic boundary principle

7.1 Three layers of entropy

Across all domains, the entropic boundary E appears in at least three compatible layers:

- **Analytic entropy:** energy dissipation, Fourier decay, regularity.
- **Algebraic entropy:** bounds on moments, enstrophy, or singular sets.
- **Generative/systemic entropy:** statistical stationarity, regeneration of coherent structures.

In Navier–Stokes, these correspond to the energy identity, enstrophy bounds, and long-term statistical behavior. In recurrences, the integer lattice plays the role of an algebraic entropic boundary. In Makeya and QM, dimensional obstructions and uncertainty act as entropic closures.

8 Program for formalization

The dm^3 Criticality Principle and its instantiations can be encoded in a proof assistant (e.g. Lean) as:

- abstract predicates for rigid, critical, and supercritical regimes,
- a conjectural axiom stating the existence of a critical value 3,
- schematic theorems asserting that Ricci 3D, NS 3D, Makeya in $\mathbb{R}^3$, quantum directional coherence, and Tribonacci all instantiate the critical regime,
- schematic theorems asserting that Tetranacci, Pentanacci, and higher recurrences instantiate supercritical regimes.

The open problems (e.g. NS regularity, full Makeya structure, Collatz) appear as *honest admits*: named conjectures with no false theorems.

9 Conclusion

We have proposed the dm^3 Criticality Principle as a unifying structural lens on a wide range of dynamical systems. The number 3 emerges as a critical curvature dimension where generative behavior becomes possible but remains controllable, with an entropic boundary enforcing closure. The recurrence ladder, Ricci flow, Navier–Stokes, Keakeya, and quantum directional coherence all align with this spine.

The next steps are:

- to refine the formal predicates for C, K, F, U, E in each domain,
- to push the formalization in a proof assistant,
- to test the principle against further domains (e.g. number theory, coding theory, statistical mechanics).

The aim is not to reduce these fields to a single slogan, but to expose a shared generative geometry that might guide both conceptual understanding and formal verification.

References

- [1] T. Wang and J. Zahl. On the dimension of Keakeya sets in $\mathbb{R}^3$. 2025.
- [2] L. Caffarelli, R. Kohn, and L. Nirenberg. Partial regularity of suitable weak solutions of the Navier–Stokes equations. *Comm. Pure Appl. Math.*, 35(6):771–831, 1982.
- [3] G. Perelman. The entropy formula for the Ricci flow and its geometric applications. 2002.